MySQL CRUD Practice – Product Table

Task: Perform CRUD operations in MySQL using a products table.

1. Create Database and Table

Create a database named product_db and a table named products with the following columns.

Column Name	Data Type	Description
id	INT	Primary key, auto increment
name	VARCHAR(100)	Product name
brand	VARCHAR(50)	Brand name
category	VARCHAR(50)	Product category
price	DECIMAL(10,2)	Product price
quantity	INT	Available stock

2. Reference Data

Insert the following 10 products into the table.

ID	Name	Brand	Category	Price (₹)	Quantity
1	Galaxy S24	Samsung	Mobile	75000	10
2	iPhone 15	Apple	Mobile	65000	8
3	IdeaPad Slim 3	Lenovo	Laptop	45000	12
4	Pavilion 15	HP	Laptop	58000	5
5	Noise ColorFit	Noise	Watch	2500	20
6	Galaxy Watch 6	Samsung	Watch	22000	7
7	Air Max	Nike	Shoes	8500	15
8	Superstar	Adidas	Shoes	6000	18
9	iPad 10th Gen	Apple	Tablet	40000	6
10	Redmi Pad SE	Xiaomi	Tablet	18000	9

3. CRUD Questions

A. Create (INSERT)

1. Insert a new product: OnePlus 12, Brand: OnePlus, Category: Mobile, Price: ₹64,999, Quantity: 10.
2. Insert two new products in a single SQL query.
3. Insert a product with a price of ₹3,500 and quantity of 25.
4. Insert a new laptop from Dell with a price of ₹55,000 and quantity of 7.
5. Insert a new product without specifying the id.

B. Read (SELECT)

6. Display all products from the table.
7. Display only the product name, brand, and price.
8. Display all products belonging to the Mobile category.
9. Find products with a price greater than ₹20,000.
10. Find products with a price between ₹5,000 and ₹40,000.
11. Display products whose quantity is less than 10.
12. Find all products manufactured by Samsung.
13. Display products in ascending order of price.
14. Display the five most expensive products.
15. Find products whose names contain the word Galaxy.
16. Count the total number of products.
17. Find the highest-priced and lowest-priced products.
18. Calculate the total stock quantity of all products.
19. Display products belonging to either the Laptop or Tablet category.
20. Display products with a price greater than ₹10,000 and quantity greater than 5.

C. Update (UPDATE)

21. Change the price of Galaxy S24 to ₹72,000.
22. Update the quantity of iPhone 15 to 15.
23. Change the brand of product ID 5 to Noise.
24. Increase the price of all laptops by 10%.
25. Reduce the quantity of all products in the Shoes category by 2.
26. Change the category of product ID 9 to Electronics.
27. Update the price to ₹19,999 for products whose price is between ₹18,000 and ₹22,000.
28. Set the quantity to 0 for products whose price is greater than ₹70,000.

D. Delete (DELETE)

29. Delete the product with ID 10.
30. Delete all products belonging to the Watch category.
31. Delete products whose quantity is 0.
32. Delete products with a price below ₹3,000.
33. Delete all products manufactured by Adidas.
34. Delete a product using its name instead of its ID.
35. Delete all records from the products table without dropping the table.